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- Welcome to Machine Learning Essentials

Machine Learning Essentials Overview

Lesson 1: Introduction to Statistical Learning

Lesson 2: Curse of Dimensionality

Lesson 3: Statistical Models and Model Quality

 Video: Parametric Models

4 min

 Video: Model Quality

6 min

 Video: Bias-Variance Tradeoff

8 min

 Graded Assignment: Learning Check - Bias-Variance Tradeoff

Submitted

 Reading: Opt-in to Penn Engineering Online Communications

1 min

Weekly Assignment

Your grade: 100%

Next item 

Your latest: 100% • Your highest: 100% • To pass you need a total of 80%. Will keep your highest score.

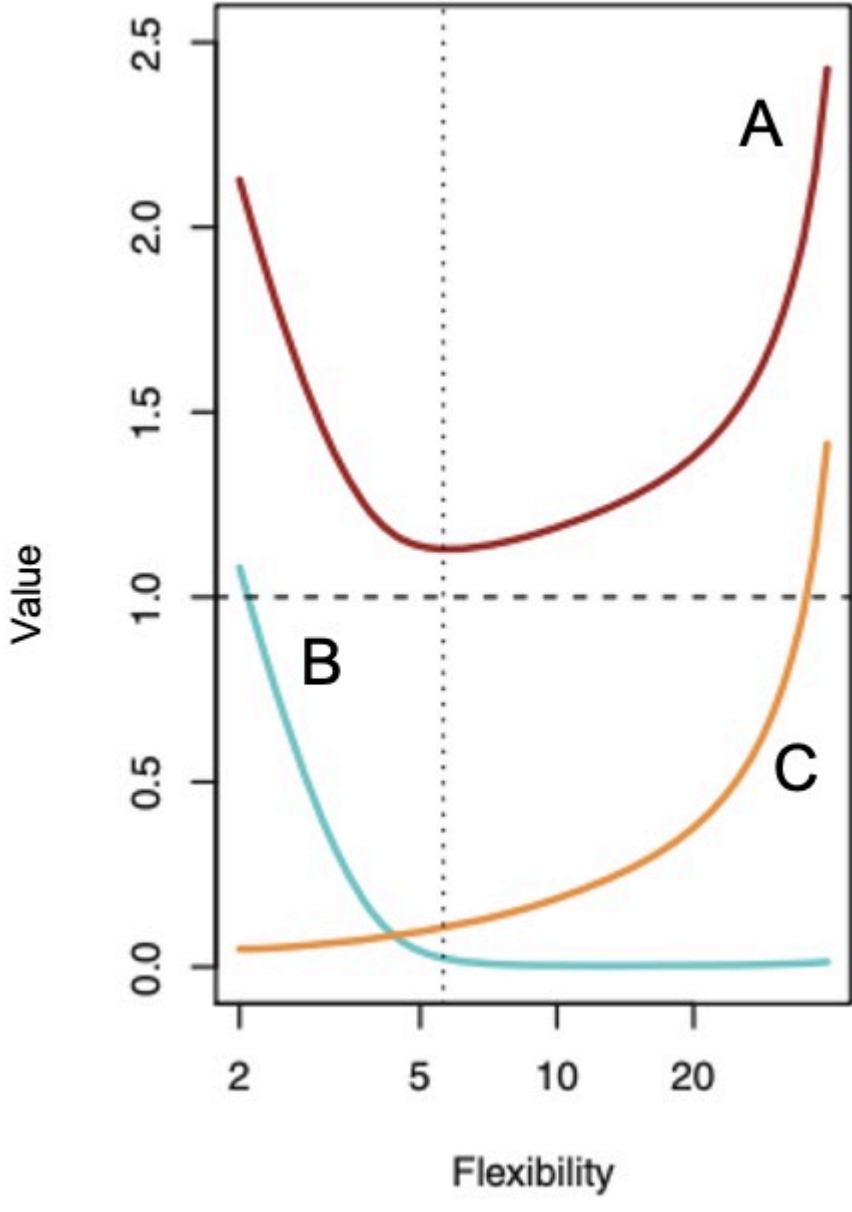
Learning Check - Bias-Variance Tradeoff

 Instructions

Review Learning Objectives

1. For Questions 1-3, use the graph shown below to indicate what function corresponds to each label. 1 / 1 point 

Points to consider about your first learning history starting the assignment? I'm here to help.



4 hours

est score.

View submission

See feedback

Try again

Line A is...

☐ Squared Bias

☐ Variance

☒ Test MSE

 Correct

Line A represents the test MSE as a function of the flexibility of our model. This function presents a minimum value at the optimal amount of flexibility.

2. Line B is...

☒ Squared Bias

☐ Variance

☐ Test MSE

 Correct

Line B indicates the squared bias. As model flexibility increases, bias will decrease, since our model will be able to track the "middle" of the cloud of points in our data set. Intuitively, a model will have high bias is it is not able to approximate to the "middle" of the cloud of points.

3. Line C is...

☐ Squared Bias

☒ Variance

☐ Test MSE

 Correct

Line C indicates the variance.

As model flexibility increases, variance will increase. Intuitively, as we increase the flexibility of our model, it will become more "wiggly", inducing a higher variance in our estimates.